

BUS713 Financial Analytics: Workshop 2

Risk, Volatility, and Value at Risk in R

Builds on Module 1: the same KO and PEP data, now analysed for risk

Module 1 answered how much you gained or lost. Module 2 answers how uncertain that outcome was. We continue with Coca-Cola (KO) and PepsiCo (PEP) rather than introducing new tickers, so every result builds on data you already understand. Work through each problem yourself, in order.

Task 1: Why Risk Is Measured on Returns, Not Prices

Scenario: *You're a risk analyst. A colleague proposes measuring Coca-Cola's risk as the standard deviation of its raw share price series. Your manager asks you to check whether that approach is sound before it goes into a report.*

A statistic like standard deviation only means what we think it means if the thing it is measuring has a stable average and stable spread over the period studied. Stock prices typically drift or trend over time, so their average level keeps changing. Returns, by contrast, tend to fluctuate around a fairly stable average. This is why financial risk is measured on returns, not prices.

1. Download KO price history and plot the adjusted price over time. Then compute its monthly returns and plot those. Compare the two charts: does the price series drift over the decades, or stay at a stable level? Does the return series behave the same way?
2. A colleague argues that "KO's price has climbed steadily for two decades, so it must be low risk." Using what you observed in the two charts, explain in 2 to 3 sentences why this reasoning is flawed.
3. Why does it matter, for calculating a meaningful standard deviation, whether the average level of a series stays stable over time? Answer with specific reference to what would happen if you calculated the standard deviation of KO's raw price instead of its returns.

Task 2: Comparing Volatility Across Assets

Scenario: *A portfolio manager wants to know whether Coca-Cola or PepsiCo carries more risk before deciding how to allocate between them.*

Volatility is the standard deviation of returns: it measures how widely returns are dispersed around their average, not whether they are positive or negative.

4. Compute PEP's monthly returns using the same approach as Task 1 for KO.
5. Calculate the mean and standard deviation of monthly returns for both stocks. Which has the higher average return? Which has the higher volatility?
6. Combine both stocks' summary statistics into one table and build a bar chart comparing their volatility with `geom_col()`. Does the stock with the higher return also carry the higher risk? What would you tell the portfolio manager?

Task 3: Detecting Volatility Clustering

Scenario: *Your manager is sceptical that risk changes over time and wants evidence before that claim goes into a client report.*

Volatility clustering means turbulent periods tend to be followed by more turbulence, and calm periods by more calm. Squared returns are a simple proxy for volatility because they capture the size of a move regardless of direction.

7. Download KO daily prices and compute daily returns. Plot the series. Identify one period that looks calm and one that looks turbulent.
8. Group the daily returns by month (`floor_date()` and `group_by()`) and calculate the standard deviation within each month to get a monthly realised volatility series. Plot it. Do high-volatility months cluster together, or look scattered?
9. Square the daily returns and plot the squared series. Use `Acf()` from the forecast package on the squared returns. Do the autocorrelations stay positive across many lags, or drop to zero quickly? What does that imply about whether volatility is persistent?

Task 4: Value at Risk

Scenario: *You're asked to quantify how much a KO position could lose on a single bad day, for a firm's daily risk report.*

Historical VaR at a given confidence level is a lower quantile of the return distribution: `quantile(returns, 0.05)` is the 95% VaR.

10. Using KO's daily returns, calculate the 5th percentile with `quantile()`. This is the 95% historical VaR.

11. A colleague holds a \$200 million position in KO. Convert the VaR into dollar terms and state it as a one-sentence risk statement a non-technical manager could understand.
12. Recalculate the 95% VaR using only the most recent two years of data instead of the full history. Does it change much? What does that imply about choosing a long historical window versus a recent one?

Task 5: Expected Shortfall and Tail Risk

Scenario: *The risk committee asks: if we do have a bad day beyond VaR, how bad could it realistically get?*

Expected Shortfall is the average return conditional on being worse than the VaR threshold: it describes severity, where VaR only describes a cutoff.

13. Using the VaR threshold from Task 4, filter KO's daily returns to the worst 5% and calculate their average. This is the 95% Expected Shortfall.
14. Compare ES to VaR. Which is more negative, and why does that follow from how each is defined?
15. Plot a histogram of KO's daily returns with a vertical line marking the VaR threshold. In 2 to 3 sentences, explain to the risk committee what ES adds beyond VaR that would matter in a genuine crisis.

Task 6: Full Risk Comparison and Recommendation

Scenario: *You're finishing the risk briefing on KO and PEP for the same client from Module 1, who is now weighing growth against downside risk before choosing between the two.*

16. Compute daily returns for PEP. Calculate its mean return, standard deviation, 95% VaR, and 95% Expected Shortfall, using the same steps as Tasks 3 to 5 for KO.
17. Combine all four statistics for both stocks into a single summary table.
18. Write a 4 to 6 sentence risk briefing recommending KO or PEP for a risk-averse client, citing the specific figures you calculated and, if useful, the CAGR figures from Module 1.

Wrap-Up

You examined why risk is measured on returns rather than prices, measured and compared volatility across KO and PEP, detected volatility clustering in real data, and calculated Value at Risk and Expected Shortfall from historical returns. Keep your scripts; the next module builds on the same return series.